

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

ADVANCED MICROPROCESSORS

MODEL QUESTION PAPER – SET-1

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

Module Outcome Cognitive level

1	8086 microprocessor has ----- address lines.	M1.01	R
2	How the minimum and maximum modes of 8086 are selected?	M1.04	R
3	What is meant by addressing mode of an instruction?	M2.02	R
4	Which one gets higher priority? Interrupt request on INTR pin or a software interrupt?	M2.01	R
5	How the two instructions MOV AX,BX and MOV AX,[BX] differ on execution?	M2.03	U
6	80386 microprocessor has ----- virtual memory.	M3.01	R
7	Why Pentium is called a superscalar processor?	M3.03	R
8	Define core.	M4.02	R
9	Name the Intel technology that allows a single processor to run as two logical processors?	M4.01	R

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

Module Outcome Cognitive level

1	List any six features of 8086.	M1.01	R
2	Describe the segment registers of 8086.	M1.03	R
3	The CS register and IP register in an 8086 contains 2435H and 167AH respectively. Explain how the 20-bit physical address is generated and how much is it in this case.	M1.02	U
4	Explain how the Queue speeds up program execution.	M1.01	U
5	State the following pin-functions of 8086; ALE, $\overline{DEN}$, DT/ $\overline{R}$	M1.04	R
6	Explain how 8086 responds to interrupts.	M2.01	U
7	Describe the following assembler directives: i) SEGMENT and ENDS ii) PROC and ENDP	M2.04	R

8	Explain virtual 8086 mode of 80386.	M3.01	U
9	Outline the technological features of IA processors.	M4.05	R
10	State any three major issues in multicore processing.	M4.03	R

PART C

III. Answer all questions from the following (6x 7 = 42 Marks)

Module Outcome Cognitive level

1	With the help of a block diagram, explain the architecture of 8086 microprocessor.	M1.01	U
OR			
2	Explain minimum mode configuration of 8086 microprocessor.	M1.02	U
3	Explain the interrupt vector table of 8086 microprocessor.	M2.01	U
OR			
4	Explain the addressing modes of 8086 with examples.	M2.02	U
5	Write an 8086 assembly language program to divide 2A35H by 25H. The numbers are given in consecutive memory locations. Store the result in memory locations after the numbers. Use assembler directives.	M2.05	A
OR			
6	Write an 8086 assembly language program to find the average of two given 8-bit numbers. Use assembler directives.	M2.02	A
7	Explain with a block, the internal architecture of 80386 microprocessor.	M3.01	U
OR			
8	With a neat block diagram, explain the architecture of Pentium.	M3.03	U
9	Explain paging mechanism in 80386 microprocessor.	M3.02	U
OR			
10	Explain the pipeline stages in Pentium.	M3.03	U
11	Compare single core and multicore processors with general block diagrams.	M4.02	U
OR			
12	Describe the concept of hyper threading technology.	M4.01	U

ADVANCED MICROPROCESSORS

SCHEME OF EVALUATION

Max. Marks: 75

Part A

I. Answer all questions in one word or one sentence.

1	20	1	9
2	By strapping pin 33 ($\overline{MN}/\overline{MX}$) to supply (V_{CC}) or ground (GND).	1	
3	Addressing mode represents the way by which an operand is specified in an instruction.	1	
4	Software interrupt.	1	
5	The first instruction copies the content of BX to AX whereas the second one copies the content of memory location pointed by BX to AX.	1	
6	64TB	1	
7	It can simultaneously execute two instructions.	1	
8	A core is a single processing unit in a CPU. (Cores are functional units made up of computational units and caches.)	1	
9	Hyper threading technology.	1	

(9 × 1 = 9 Marks)

Part B

II. Answer any eight questions from the following, each question carries 3 marks.

1	16-bit Microprocessor ☐ 20 address lines ☐ Capable of addressing 1MB memory locations ☐ Employs two stage pipelining to speed up processing ☐ It has a 6-byte instruction Queue to store pre-fetched instruction code bytes ☐ Architecture designed for powerful assembly language and efficient high level languages ☐ Supports memory segmentation ☐ Two modes of operation i) Minimum mode ii) Maximum mode ☐ 14 word, by 16-bit register set with symmetrical operations ☐ 24 operand addressing modes ☐ Bit, Byte, Word and Block operations ☐ 8 and 16 bit signed and unsigned arithmetic in binary or decimal including multiply and divide ☐ Range of clock rates: 5MHz for 8086 8MHz for 8086-2 10MHz for 8086-1 ☐ It has 256 vectored interrupts. ☐ It consists of 29,000 transistor (Any 6)	3	
2	8086 microprocessor has four segment registers; CS, DS, SS and ES. These segment registers are used to hold the upper 16 bits of the starting address of a logical segment. This address is known as the segment base. CS or the code segment register is used to hold the		

	segment base address of code segment intended for storing the program. DS or data segment register is used to hold the segment base address of the data segment intended for storing data. SS and ES are used to hold the segment base address of stack segment and extra segment respectively. Stack segment acts as the stack memory and extra segment is used to store data during string operations.	3	24																		
3	To generate the 20-bit physical address in memory segment, the corresponding segment register content is shifted left by four bits and a 16-bit offset is added with it. For code segment, the segment base address is there in CS register and the offset is held by IP register. <table><tr><td>CS</td><td>2</td><td>4</td><td>3</td><td>5</td><td></td></tr><tr><td>IP</td><td></td><td>1</td><td>6</td><td>7</td><td>A</td></tr><tr><td>Physical Address</td><td>2</td><td>5</td><td>9</td><td>C</td><td>A</td></tr></table>	CS		2	4	3	5		IP		1	6	7	A	Physical Address	2	5	9	C	A	3
CS	2	4		3	5																
IP		1		6	7	A															
Physical Address	2	5		9	C	A															
4	- Queue is a six byte long first-in-first-out(FIFO) register array - Used to store up to six bytes of pre-fetched instructions for the EU - When the EU is ready for the next instruction, it is simply taken from the Queue, rather than sending out the address and waiting for the memory to supply the code bytes. Thus the Queue speeds up processing	3																			
5	ALE: Address Latch Enable. Outputs a pulse at the beginning of the bus cycle and is to indicate an address is available on the address pins. Signal is used to strobe address latches. $\overline{DEN}$: Data Enable. Output during the latter portion of the bus cycle and is to inform the transceivers that the CPU is ready to send or receive data. Signal is used to enable data transceiver IC. DT/$\overline{R}$: Data Transmit/Receive. Indicates to the set of transceivers whether they are to transmit (1) or receive (0) data.	3																			
6	The 8086 checks for an interrupt request at the end of each instruction cycle. If an interrupt has been requested, 8086 responds to the interrupt by carrying out the following steps: 1. It decrements the stack pointer by 2 and pushes the flag register on to the stack. 2. It resets the interrupt flag (IF) in the flag register. 3. It resets the trap flag (TF) in the flag register. 4. It decrements the stack pointer by 2 and pushes the current code segment register contents on to the stack. 5. It decrements the stack pointer again by 2 and pushes the instruction pointer contents on to the stack. 6. It does an indirect far jump to the start of the procedure that has been written to respond to the interrupt.	3																			
7	- SEGMENT directive is used to indicate the start of a logical segment. The segment directive is preceded by the name of the segment. The statement, CODE SEGMENT, indicates to the assembler the start of a logical segment called CODE. - ENDS directive indicates the end of a logical segment. The SEGMENT and ENDS directives are used to bracket a logical segment containing code or data. - PROC directive is used to indicate the start of a procedure. The	3																			

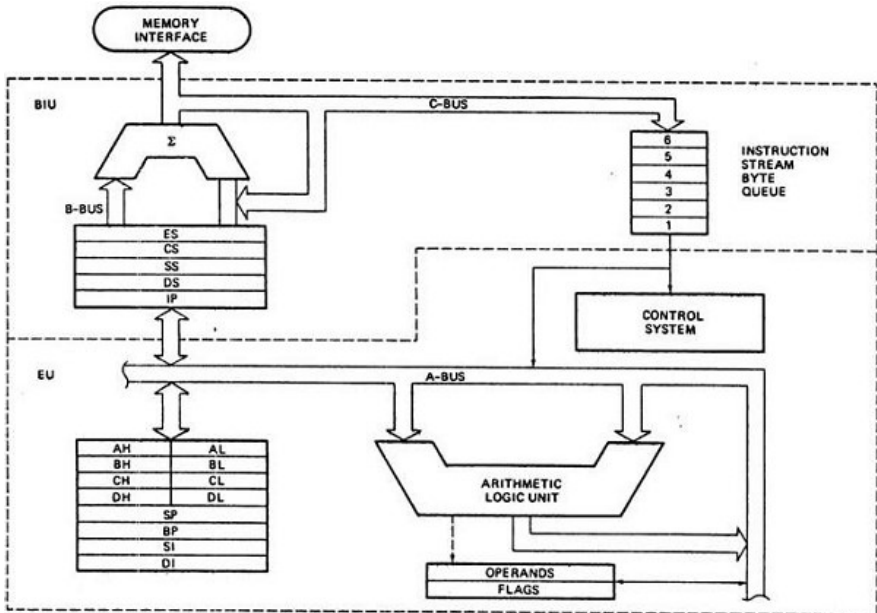
	PROC directive is preceded by the name of the procedure. The PROC directive may be followed by the term near or the term far to specify the type of the procedure. • ENDP indicates the end of a procedure. The PROC directive is used with the ENDP directive to bracket a procedure.		
8	• The virtual 8086 mode permits the execution of 8086 applications with all protection features of 80386. • In V86 mode the segment registers are used similar to that of real mode. • In this mode the processor computes 20-bit address by shifting the segment register left by four times and adding to offset. • The 20-bit address can be used to access 1Mb of physical memory space. • In virtual 8086 mode, paging can be enabled to run multiple virtual mode tasks, to provide protection and operating system isolation. • When paging is enabled, the 20-bit linear address can be divided into 256 pages and each page can be located anywhere in the 4Gb physical address space of 80386.	3	
9	i) Virtualization: Intel's Virtualization technology allows systems to work with more than one operating system. ii) Turbo Boost: Intel's Turbo Boost technology allows the processor core to run automatically at a higher clock frequency if it is operating within its temperature, power and current limits. iii) Hyperthreading: Intel's Hyperthreading technology allows a single processor to run as two logical processors. iv) Advanced vector extensions: This extension of the IA instruction set uses 256-bit vector data to perform operations of integer, floating point, fused multiply add instructions, etc. This is useful for math and DSP operations.	3	
10	i) Interconnection: With multiple cores and shared resources such as memory, buses, and I/O, the interconnection between cores is a big issue. Bus contention and interprocessor communication are the main problems in any multiprocessing systems which is the same for multicore processors. ii) Cache Coherence: Cache coherence is an issue in multicore processors because of the distributed private caches and shared main memory. In most multicore processors, the L1-caches are private and the L2-caches are shared. Since each core has its own private cache, the copy of the data in that cache may not be the up-to-date version. For ensuring cache coherence, two protocols; snooping protocol and directory based protocols are used. iii) Programming aspects: Exploiting a multicore processor to its full potential is a big issue. Intelligent programming with multiple thread possibilities or parallelization is	3	

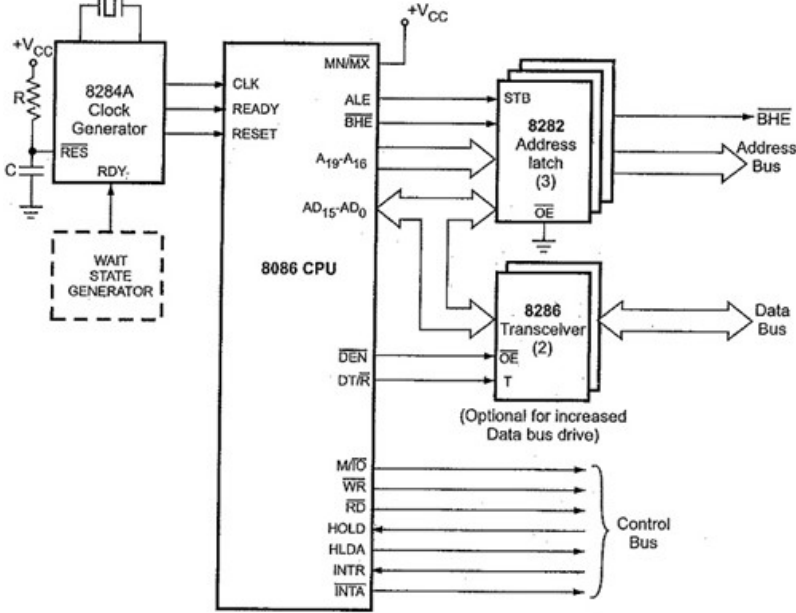
	essential to ensure simultaneous execution in all the cores and thus fully exploiting the potentials of multicore processor.		
iv)	Operating System Support: The operating system that used with a multicore processor must be able to handle the complexities of such a system to fully exploit the available resources.		
	(Any three)		

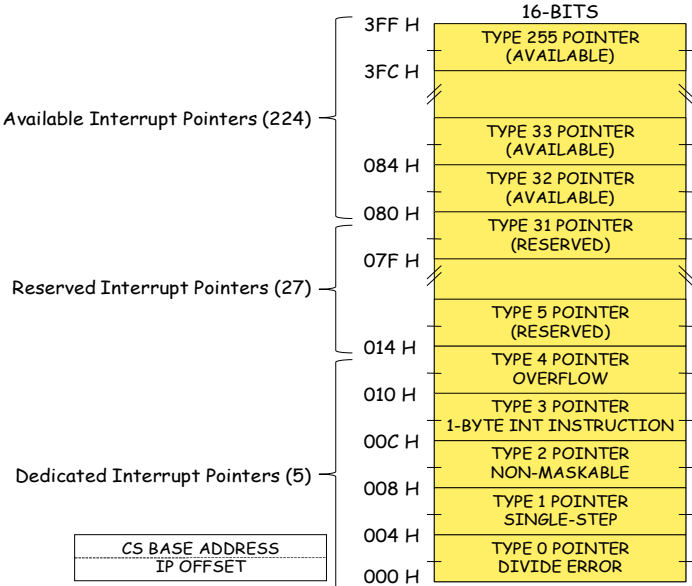
(8 × 3 = 24 Marks)

Part C

III. Answer all questions. Each question carries 7 marks.

1	 8086 Microprocessor is divided into two functional units, i.e., EU (Execution Unit) and BIU (Bus Interface Unit). EU (Execution Unit) Execution unit gives instructions to BIU stating from where to fetch the instructions and then decode and execute those instructions. Its function is to control operations on data using the instruction decoder & ALU. ALU: It handles all arithmetic and logical operations. Flag Register: It is a 16-bit register that behaves like a flip-flop. It changes its status according to the result stored in the accumulator. It has 9 flags and they are divided into 2 groups – 6 Conditional Flags; CF, PF, AF, ZF, SF, OF and 3 Control Flags; TF, IF, DF. General purpose register: There are 8 general purpose registers, i.e., AH, AL, BH, BL, CH, CL, DH, and DL. These registers can be used individually to store 8-bit data and can be used in pairs to store 16bit data. Stack pointer register: It is a 16-bit register, which holds the address from the start of the segment to the memory location, where a word was most recently stored on the stack. BIU (Bus Interface Unit) BIU takes care of all data and addresses transfers on the buses for the EU like	Fig-4 Exp-3
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	sending addresses, fetching instructions from the memory, reading data from the ports and the memory as well as writing data to the ports and the memory. It has the following functional parts; Instruction queue: A 6-byte long register array that works in first in first out manner. It is used to store pre-fetched instructions. Segment registers – BIU has 4 segment registers, i.e. CS, DS, SS & ES. They hold the starting addresses of memory segments used to store instructions and data in memory. Instruction pointer – It is a 16-bit register used to hold the address of the next instruction to be executed.	
OR		
2	 • In minimum mode configuration the MN/MX pin is strapped to +5V • In this mode, the microprocessor directly generates all the control and status signals • All the address lines are multiplexed and so they carry the address bits only during the first part of the bus cycle. Three 8282 latches are used to store these bits for further use • The BHE signal also has to be latched • The ALE signal is used to enable these latches • Two transceiver ICs (8286) are used to control the data flow through the data bus • The transceivers also increase the current sourcing/sinking capacities • The transceivers are enabled by the DEN signal and the direction of data flow is controlled by the DT/R signal • A clock generator, 8284A, is used in the system to supply clock pulses as well as to synchronize the READY and RESET signals	Fig-4 Exp-3

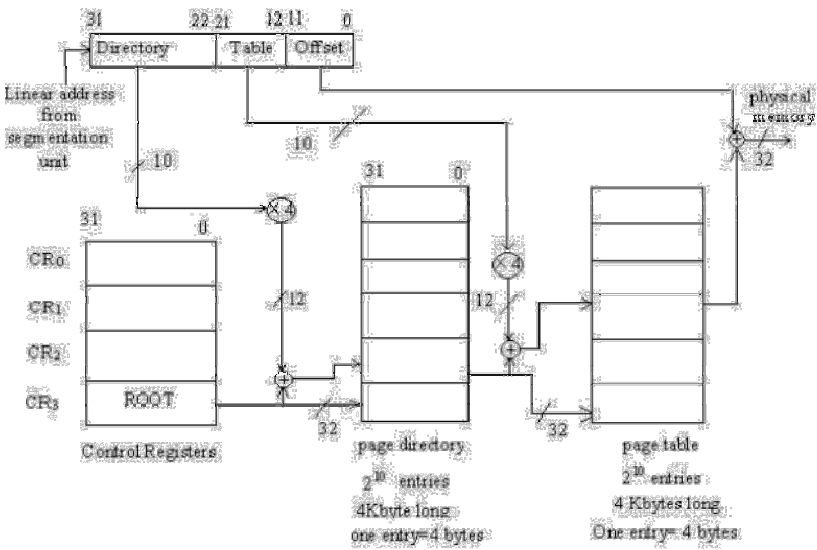
3	 Available Interrupt Pointers (224) Reserved Interrupt Pointers (27) Dedicated Interrupt Pointers (5) CS BASE ADDRESS IP OFFSET 16-BITS 3FF H TYPE 255 POINTER (AVAILABLE) 3FC H 084 H TYPE 33 POINTER (AVAILABLE) 080 H TYPE 32 POINTER (AVAILABLE) 07F H TYPE 31 POINTER (RESERVED) 014 H TYPE 5 POINTER (RESERVED) 010 H TYPE 4 POINTER OVERFLOW 00C H TYPE 3 POINTER 1-BYTE INT INSTRUCTION 008 H TYPE 2 POINTER NON-MASKABLE 004 H TYPE 1 POINTER SINGLE-STEP 000 H TYPE 0 POINTER DIVIDE ERROR In an 8086 system, the first 1K Bytes of memory, from 00000H to 003FF H, is set aside as a table for storing the starting addresses of interrupt service procedures. Each procedure requires four bytes for storing the CS and IP values, so the table can hold the starting address of up to 256 interrupt procedures. The starting address of an interrupt service procedure is called the interrupt vector or the interrupt pointer, so the table is referred to as the interrupt vector table or the interrupt-pointer table. The interrupt vector table is shown in figure. In each vector, the IP value is stored as the low word and the CS value is stored as the high word. Each double word interrupt vector is identified by a number from 0 to 255. This number is called the type of the interrupt. The lowest five types are dedicated interrupts. Interrupt types 5 to 31 are reserved by Intel for use in more complex microprocessors. The upper 224 interrupt types from 32 to 255 are available to be used for hardware or software interrupts.	Fig-4 Exp-3
4	OR Register Addressing Mode When the name of a register is used to identify the location of data, the addressing mode is called register addressing mode. Eg. MOV AH, BL Immediate Addressing Mode In this mode, 8-bit or 16-bit datum is directly given in the instruction. Eg. MOV AL, 05 H Direct Addressing Mode In this addressing mode, the 16-bit effective address (offset) of the datum is given as part of the instruction Eg. MOV AL, [8200 H] Indirect Addressing Mode In this mode, the effective address of the datum is not directly given in the instruction. Instead, it may be given in a base or index register, or	7

	given as the sum of a base and index register, or as the sum of a displacement (8-bit or 16-bit) and a base or/and index register. Based or Indexed Addressing Mode In this addressing mode, the effective address of the datum is given in a base register/index register specified in the instruction. eg. MOV AL, [BX] MOV AL, [SI] Relative Based or Indexed Addressing Mode Here, the effective address of the datum is obtained by adding the content of a specified base or index register and a 8-bit or 16-bit displacement given in the displacement field of the instruction. eg. MOV AL, PRICE [BX] MOV AX, VALUE [DI] Based Indexed Addressing Mode In this addressing mode, the effective address of the datum is given as the sum of the contents of a base register and an index register. eg. MOV AL, [BX][SI] Relative Based Indexed Addressing Mode In this mode, the effective address of the operand is obtained by adding the content of a base register, an index register and a displacement (8-bit or 16-bit) specified in the instruction eg. MOV AL, PRICE [BX][SI]. Implied Addressing Mode Instruction using this addressing mode has no operands. eg. DAA	7	
5	<pre> DATA SEGMENT DIVIDEND DW 2A35H DIVISOR DB 25H QUOTIENT DB 1DUP(0) REMAINDER DB 1DUP(0) DATA ENDS CODE SEGMENT ASSUME CS:CODE,DS:DATA MOV AX,DATA MOV DS,AX MOV AX,DIVIDEND DIV DIVISOR MOV QUOTIENT,AL MOV REMAINDER,AH CODE ENDS END </pre>	7	
OR			
6	<pre> DATA SEGMENT NUM_1 DB 25H NUM_2 DB 3AH AV_NUM DB 1DUP(0) DATA ENDS CODE SEGMENT ASSUME CS:CODE,DS:DATA MOV AX,DATA MOV DS,AX MOV AL,NUM_1 ADD AL,NUM_2 </pre>	7	

	<pre> MOV AH,00H ADC AH,00H MOV BL,02H DIV BL MOV AV_NUM,AL CODE ENDS END </pre>		
7	80386 has highly pipelined architecture with six functional units operating in parallel. The six functional units are: ▪ Bus Interface Unit ▪ Instruction (Code) Prefetch Unit ▪ Instruction Decode Unit ▪ Execution Unit ▪ Segmentation Unit ▪ Paging Unit Bus Interface Unit: Controls all the transfer of address and data over the buses. It generates control signals for memory and I/O interface. Instruction Prefetch Unit: This unit fetches instructions in advance and stores them in the 16-byte code queue, whenever the bus is free. Instruction Decode Unit: Reads the instructions from prefetch unit and decodes them. The decoded instructions are then stored in a 3-byte queue. Execution Unit: Read the decoded instructions from decode unit and process them under the control of microcode stored in a control ROM and execute them with the help of a data unit. Segmentation and Paging Units: These units act as the Memory Management Unit (MMU) of 80386. In real address mode, paging is disabled and the segmentation unit computes the address similar to that of 8086 or 80286 in real mode. In protected virtual address mode (PVAM) the segmentation unit computes a 32-bit linear address from a 48-bit virtual address. If paging unit is not enabled, this linear address is used as the physical address. If paging is enabled, then the paging unit will translate the 32-bit linear address to 32-bit physical address. The paging provides an additional		

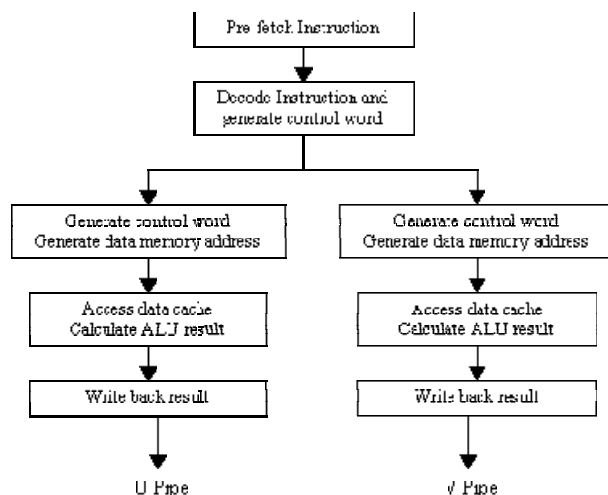
Fig-4
Exp-3

	memory management mechanism to handle very large segments.	7
8	The Pentium processor has superscalar architecture which allows parallel execution of two instructions. The various functional blocks of Pentium processor are shown in figure. The bus unit takes care of issuing control signals and fetching code/data from external memory or I/O devices. Pentium processor employs two separate cache memories each 8Kb in size; one for code and one for data. The code cache is connected to prefetch buffer by 256-bit bus. Hence in one clock cycle, the instruction cache can provide upto 32 bytes of raw opcodes to prefetch buffer. The data cache has two ports and each port is connected to pipelines by 32-bits. Hence the data cache can provide two data simultaneously. The TLBs associated with the caches translates linear addresses to physical addresses used by the cache. Pentium processor has two 32-byte prefetch buffers. Only one prefetch buffer is active at any one time. One buffer prefetches instructions sequentially and the other one starts prefetching from a branch target address when the Branch Target Buffer predicts that a branching will take place. The execution unit has two parallel instruction pipelines, the "u" pipe and the "v" pipe with individual ALU for each pipe. The u-pipe can execute all integer and floating point instructions. The v-pipe can execute simple integer instructions and the FXCH floating-point instructions. The decode unit decodes the pre-fetched instructions so the Pentium processors can execute the instruction. The control ROM contains the microcode which controls the sequence of operations that must be performed to execute an instruction. The control ROM unit has direct control over both pipelines. The Pentium processors contain a pipelined floating-point unit	Fig-4 Exp-3

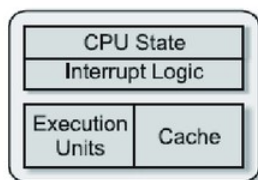
	that is ten times faster than the one used in 80486.		
9	 The diagram illustrates the x86-64 paging mechanism. A 32-bit linear address is divided into three parts: a 10-bit Directory field, a 10-bit Table field, and a 12-bit Offset field. The Directory field is used to select one of the four Control Registers (CR0, CR1, CR2, CR3). CR3 contains the 'ROOT' address, which is the base address of the page directory. The Table field is used to select one of the 2^10 (1024) entries in the page directory. Each entry in the page directory is 4 bytes long. The selected entry points to a page table. The Offset field is used to select one of the 2^10 (1024) entries in the page table. Each entry in the page table is 4 bytes long. The selected entry in the page table provides the upper 20 bits of the physical address. The 12-bit Offset is then concatenated with these 20 bits to form the final 32-bit physical address. • The paging provides a method for managing very large memory segments defined in PVAM. The paging divides each segment into equal sized (4Kb) pages. • When paging is enabled, the segmentation unit will generate a 32-bit linear address and supply to paging unit, which translates this linear address to 32-bit physical address. • The linear address has three components; a 10-bit directory index, 10-bit page table index and 12-bit offset. • The control register CR3 holds the base/root address of page directory. • The root address is added to 10-bit page directory index (multiplied by 4) to get the address of page directory entry of the page table. • The page directory entry has the base address of page table, which is added to 10-bit page table index (multiplied by 4) to get the address of page table entry of the page. • The page table entry has the upper 20-bits of the page frame address, which is concatenated with lower 12-bits of the linear address to form the physical address.	Fig-4 Exp-3	
OR			
10	Each of the two pipelines has five stages 1] Prefetch : Instructions are fetched from the instruction cache and aligned in pre-fetch buffers for decoding. 2] D1 instruction decode : Two parallel decoder works together to decode and generate control signals.		7

- 3] D2 Address generate :By decoding the addresses of memory resident operands are calculated.
- 4] Execute: The integer hardware executes the instruction.
- 5] Write Back : In this stage the contents of registers and status flags are updated depending upon instructions..

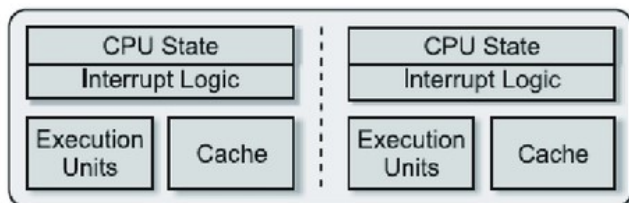
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11



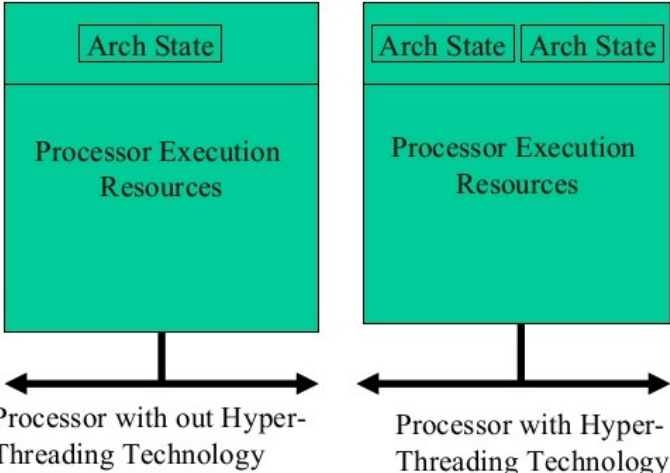
(a) Single core



(c) Multi-core

Parameter	Single Core	Multicore
No. of Cores	One primary core	Two or more separate cores
Processing	Sequential	Parallel
SMT	Not possible	Possible
Power	Low	High
Speed	Slow	Fast
Efficiency	Low	High

Fig-3
Comp-4

	<table> <tr> <td><i>Operation</i></td><td><i>One task at a time</i></td><td><i>Multitasking</i></td></tr> <tr> <td><i>Examples</i></td><td><i>8086,80386,Pentium I,II,III</i></td><td><i>Core 2 Duo, i3,i5,i7</i></td></tr> </table>	<i>Operation</i>	<i>One task at a time</i>	<i>Multitasking</i>	<i>Examples</i>	<i>8086,80386,Pentium I,II,III</i>	<i>Core 2 Duo, i3,i5,i7</i>	
<i>Operation</i>	<i>One task at a time</i>	<i>Multitasking</i>						
<i>Examples</i>	<i>8086,80386,Pentium I,II,III</i>	<i>Core 2 Duo, i3,i5,i7</i>						
	OR							
12	 • Hyper-Threading is a technology used by some Intel microprocessors that allows a single microprocessor to act like two separate processors • Hyper-threading is used to improve parallelization of computations (doing multiple tasks at once) performed on PC microprocessors. • For each processor core that is physically present, the operating system addresses two virtual processors, and shares the workload between them when possible. • Hyper-threading works by duplicating certain sections of the processor—those that store the architectural state—but not duplicating the main execution resources. • A hyper-threading processor appear as two "logical" processors to the host operating system, allowing the operating system to schedule two threads or processes simultaneously. • When execution resources would not be used by the current task in a processor without hyper-threading, and especially when the processor is stalled, a hyper-threading equipped processor can use those execution resources to execute another scheduled task. • This technology is transparent to operating systems and programs. The minimum that is required to take advantage of hyper-threading is symmetric multiprocessing (SMP) support in the operating system, as the logical processors appear as standard separate processors. • This technology has been implemented in Intel's Atom, Core i3, Core i5, Core i7, Itanium, Pentium 4 and Xeon CPUs.	7						

(6 × 7 = 42 Marks)

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FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

ADVANCED MICROPROCESSORS

MODEL QUESTION PAPER – SET-2

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

Module Outcome Cognitive level

1	Define pipelining.	M1.01	R
2	If the 8086 microprocessor has to be worked in a multiprocessor system, which mode is to be selected?	M1.04	U
3	What is an assembler?	M2.02	R
4	Give the type number associated with non maskable interrupt.	M2.01	R
5	State the purpose of ESC instruction.	M2.03	R
6	Which operating mode permits 80386 to run 8086 programs with full protection?	M3.01	R
7	What is meant by bus cycle pipelining?	M3.03	R
8	What is a homogeneous multicore processor.	M4.02	R
9	What is a thread?	M4.01	R

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

Module Outcome Cognitive level

1	Draw the bit pattern of the 8086 flag register. Brief each flag.	M1.01	U
2	Sketch and describe the general purpose registers of 8086. State the special functions of each pair.	M1.03	R
3	After a reset, from where the 8086 microprocessor start execution? Why? Illustrate the physical address generation after reset.	M1.02	U
4	State the following pin functions of 8086; $\overline{LOCK}$, $\overline{TEST}$, READY	M1.04	R
5	Classify the instruction set of 8086.	M2.03	R
6	Illustrate 8086 interrupt response.	M2.01	U

7	Define assembler directive. Explain the following assembler directives: i) ASSUME ii) ORG	M2.04	U
8	List any six features of 80386 microprocessor.	M3.01	R
9	Compare homogeneous and heterogeneous multicore processors.	M4.02	U
10	Differentiate between Core i3, i5 and i7 processors.	M4.05	U

PART C

III. Answer all questions from the following (6x 7 = 42 Marks)

Module Outcome Cognitive level

1	Draw the pin diagram of 8086 microprocessor.	M1.04	U
OR			
2	Explain memory segmentation in 8086 microprocessor with a neat sketch.	M1.02	U
3	Explain the hardware interrupts in 8086 microprocessor.	M2.01	U
OR			
4	Outline the shift and rotate instructions of 8086 microprocessor.	M2.03	U
5	Write an 8086 assembly language program to multiply FA12H by B5H. The numbers are given in consecutive memory locations. Store the result in memory locations after the numbers. Use assembler directives.	M2.05	A
OR			
6	Write an 8086 assembly language program to find the average of 'n' 8-bit numbers. Use assembler directives. The maximum value of 'n' is FF.	M2.02	A
7	Explain the real and protected virtual address modes of 80386 microprocessor.	M3.01	U
OR			
8	With a neat sketch explain the address translation in PVAM mode of 80386 without paging.	M3.02	U
9	List the main features of Pentium processor	M3.03	R
OR			
10	Draw the internal architecture of Pentium and explain how it execute two instructions simultaneously?	M3.03	U
11	Draw and explain the internal architecture of Intel Core 2 Duo.	M4.04	U
OR			
12	Explain the major issues in multicore processing.	M4.03	U

ADVANCED MICROPROCESSORS

SCHEME OF EVALUATION

Max. Marks: 75

Part A

I. Answer all questions in one word or one sentence.

1	Fetching the next instruction while the current instruction is executing is called pipelining .	1	9
2	Maximum mode.	1	
3	An assembler is a program that is used to translate the assembly language program to machine language.	1	
4	Type 2	1	
5	This instruction is used to pass instructions to a coprocessor such as 8087 math coprocessor which shares the address and data bus with 8086.	1	
6	Virtual 8086 mode.	1	
7	A feature that permits two bus cycles to be in progress simultaneously.	1	
8	Multicore processor with identical cores.	1	
9	Threads are the virtual components or codes, which divides the physical core of a CPU into virtual multiple cores. (A thread is a virtual version of a CPU core)	1	

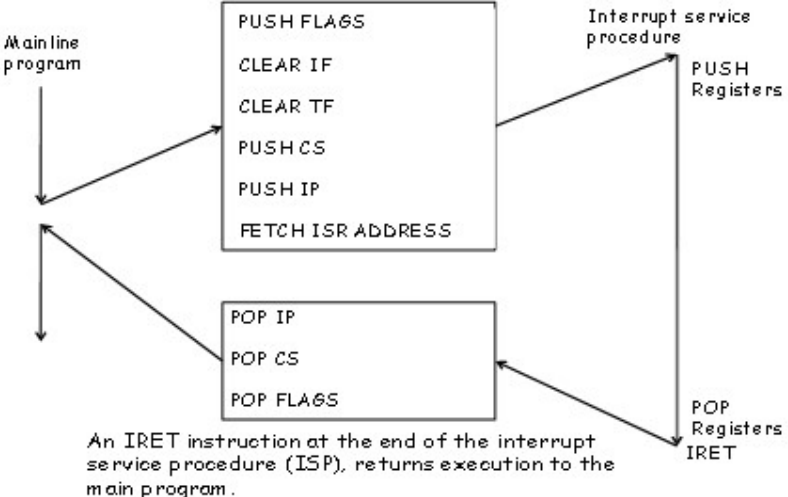
(9 × 1 = 9 Marks)

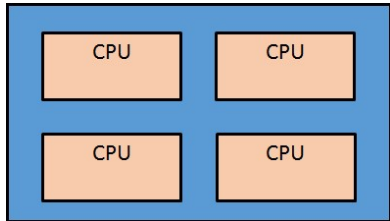
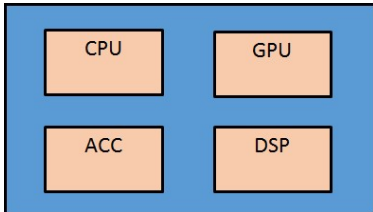
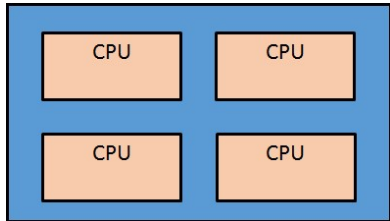
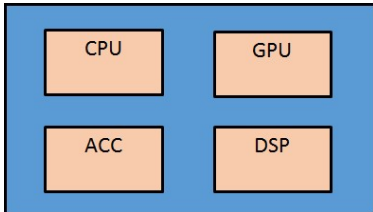
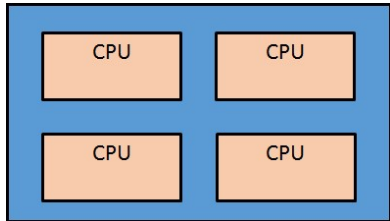
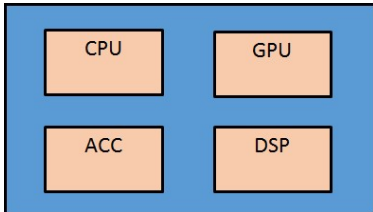
Part B

II. Answer any eight questions from the following, each question carries 3 marks.

1	8085 Compatible Flags BIT 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 U U U U OF DF IF TF SF ZF U AF U PF U CF U = Undefined Overflow String direction Interrupt enable Single step trap Sign Flag = MSB of result Zero Flag : Set if result = 0 Auxiliary carry flag for BCD Parity Flag : Set if result has even parity Carry Flag : Set by carry out of MSB	3	
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2	<table><tr><td>AH</td><td>AL</td></tr><tr><td>BH</td><td>BL</td></tr><tr><td>CH</td><td>CL</td></tr><tr><td>DH</td><td>DL</td></tr><tr><td colspan="2">SP</td></tr><tr><td colspan="2">BP</td></tr><tr><td colspan="2">SI</td></tr><tr><td colspan="2">DI</td></tr></table> There are 8 general purpose registers, i.e., AH, AL, BH, BL, CH, CL, DH, and DL. These registers can be used individually to store 8-bit data and can be used in pairs to store 16bit data. The valid register pairs are AH and AL, BH and BL, CH and CL, and DH and DL. It is referred to the AX, BX, CX, and DX respectively. • AX register – It is also known as accumulator register. It is used to store operands for arithmetic operations. • BX register – It is used as a base register. It is used to store the starting base address of the memory area within the data segment. • CX register – It is referred to as counter. It is used in loop instruction to store the loop count. • DX register – This register is used to hold I/O port address for I/O instruction.	AH	AL	BH	BL	CH	CL	DH	DL	SP		BP		SI		DI		3			
AH	AL																				
BH	BL																				
CH	CL																				
DH	DL																				
SP																					
BP																					
SI																					
DI																					
3	On RESET, the IP is cleared and CS is set to FFFF. So after a reset execution starts from FFFF0. <table><tr><td>CS</td><td>F</td><td>F</td><td>F</td><td>F</td><td></td></tr><tr><td>IP</td><td></td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>Physical Address</td><td>F</td><td>F</td><td>F</td><td>F</td><td>0</td></tr></table>	CS	F	F	F	F		IP		0	0	0	0	Physical Address	F	F	F	F	0	3	24
CS	F	F	F	F																	
IP		0	0	0	0																
Physical Address	F	F	F	F	0																
4	$\overline{LOCK}$:When this signal is active, it indicates to the other processors not to ask the CPU to leave the system bus. It is activated using the LOCK prefix on any instruction and is available at pin 29. $\overline{TEST}$: Used in conjunction with the WAIT instruction in multiprocessing environments. A WAIT instruction will cause the CPU to idle, except for processing interrupts, until a ‘0’ is applied to this pin. <i>READY</i>: Acknowledgement from memory or I/O interface that CPU can complete the current bus cycle	3																			
5	The instruction set of 8086 can be classified into six functional groups ▪ Data transfer instructions ▪ Arithmetic instructions ▪ Bit manipulation instructions ▪ String instructions ▪ Program execution transfer instructions ▪ Processor control instructions	3																			

6	 The flowchart illustrates the execution of an Interrupt Service Procedure (ISP). It begins with the 'Main line program' (represented by a downward arrow). An arrow points from the main program to a box containing the following instructions: PUSH FLAGS, CLEAR IF, CLEAR TF, PUSH CS, PUSH IP, and FETCH ISR ADDRESS. From this box, an arrow points to another box labeled 'Interrupt service procedure'. This box contains: PUSH Registers, POP Registers, and IRET. An arrow points from the 'Interrupt service procedure' box back to a third box containing: POP IP, POP CS, and POP FLAGS. Finally, an arrow points from this third box back to the 'Main line program' (represented by a downward arrow). Below the boxes, a note states: 'An IRET instruction at the end of the interrupt service procedure (ISP), returns execution to the main program.'	3	
7	The assembler directives are the instructions to the assembler regarding the program being assembled. ASSUME The ASSUME directive is used to tell the assembler the name of the logical segment it should use for a specified segment. The general form of a statement using ASSUME directive is given below ASSUME seg.reg:seg.name,seg.reg:seg.name,seg.reg:seg.name For example, the statement ASSUME CS:CODE tells the assembler that the instructions for a program are in a logical segment named CODE ORG – Originate As the assembler assembles a section of data declarations or instruction statements, it uses a location counter to keep track of how many bytes it is from the start of a segment at any time. The location counter is automatically set to 0000 when the assembler starts reading a segment. The ORG directive is used to set the location counter to a desired value at any point in the program. The statement ORG 1000h, tells the assembler to set the location counter to 1000h.	3	
8	• Uses 275,000 transistors • Highly pipelined architecture • Instruction fetching, instruction decoding, instruction execution and memory management are all carried out in parallel • 32-bit data bus • 32-bit non-multiplexed address bus • $2^{32} = 4$ Gigabyte of physical memory • 2^{46} or 64 Terabyte of virtual memory • Instruction sets of Intel microprocessors have upward compatibility • Multitasking • 80386 supports variety of operating clock frequency; 16 MHz, 20 MHz, 25 MHz and 33 MHz. (Any six)	3	
9	In a homogeneous multicore processor, all the cores are identical,		

	whereas non-identical cores constitute a heterogeneous multicore processor.																
	<table><tr><th>Homogeneous</th><th>Heterogeneous</th></tr><tr><td>Symmetric, Same Cores (Usually CPUs)</td><td>Asymmetric, Different Cores (CPUs, GPUs, DSPs and Accelerators)</td></tr><tr><td>Operation is guaranteed to be same in each core</td><td>Operation cannot be same at each core</td></tr><tr><td>Easy to offload tasks</td><td>More complicated to offload tasks</td></tr><tr><td>Good compatibility</td><td>Less compatibility</td></tr><tr><td></td><td>Specialized for specific tasks</td></tr><tr><td></td><td></td></tr></table>	Homogeneous	Heterogeneous	Symmetric, Same Cores (Usually CPUs)	Asymmetric, Different Cores (CPUs, GPUs, DSPs and Accelerators)	Operation is guaranteed to be same in each core	Operation cannot be same at each core	Easy to offload tasks	More complicated to offload tasks	Good compatibility	Less compatibility		Specialized for specific tasks			3	
Homogeneous	Heterogeneous																
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Operation is guaranteed to be same in each core	Operation cannot be same at each core																
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Good compatibility	Less compatibility																
	Specialized for specific tasks																
																	
10	1. Core i5 and i7 CPUs usually have 4 cores, while i3 CPUs only have 2. 2. Core i5 and i7 usually operate at higher frequencies than i3. 3. Both Core i5 and i7 have Turbo Boost, while Core i3 does not. 4. All CPUs have the same graphics usually, all though the speed of that graphics will depend on the individual CPU. 5. Core i3 has smaller L3 caches compared to the other two. 6. Core i5 does not support hyperthreading.	3															

(8 × 3 = 24 Marks)

Part C

Answer all questions. Each question carries 7 marks.

1			
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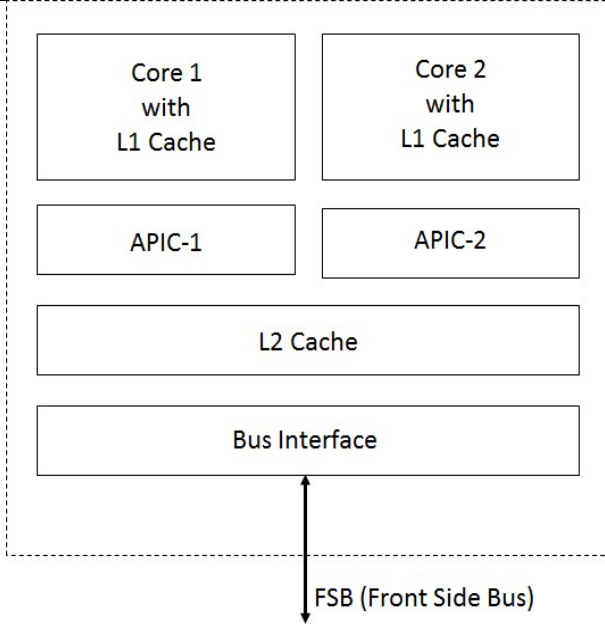
	MAX MODE MIN MODE	7
	OR	7
2	Physical Address Fig. 6.4 Memory segmentation The physical address of the 8086 Internal Architecture is 20-bits wide to access 1 Mbyte memory locations. However, its registers and memory	Fig-3 Exp-4

	locations which contain logical addresses are just 16-bits wide. Hence 8086 uses memory segmentation. It treats the 1 Mbyte of memory as divided into segments, with a maximum size of a segment as 64 Kbytes. Thus any location within the segment can be accessed using 16 bits. The 8086 Internal Architecture allows only four active segments at a time, a code segment to store the program, a data segment to store data, a stack segment to act to as the stack memory and an extra segment to store data during string operations as shown in figure. For the selection of the four active segments, four 16-bit segment registers are provided within the BIU of the 8086. These four registers are: Code segment (CS) register, the data segment (DS) register, the stack segment (SS) register, and the extra segment (ES) register. These are used to hold the upper 16-bits of the starting addresses of the four memory segments, on which 8086 works at a particular time. For example, the value in CS identifies the starting address of 64 K-byte segment known as code segment. The starting address is also known as base address or segment base. The BIU always inserts zeros for the lowest 4 bits (nibble) in the contents of segment register to generate 20-bit base address. For example, if the code segment register contains 348AH, then code segment will start at address 348A0H.		
3	Interruption caused by an external signal applied to either the NMI or the INTR pin is called a hardware interrupt. NMI (Non Maskable Interrupt) The NMI input can be used by an external hardware to interrupt an 8086 program. It is a non maskable interrupt and can be used for emergency purposes. NMI is an edge triggered interrupt and is recognized when a low to high transition occurs at this pin. It causes a type 2 interrupt. INTR Interrupts – Type 0 through 255 The INTR input can be used by an external hardware to interrupt an 8086 program. The INTR interrupt is maskable. An interrupt through INTR is effective only if the IF flag is set. If the IF flag is set, a high level on the INTR line, will cause an interrupt. As a response to this the 8086 clears IF, to prevent the 8086 from being interrupted continuously if the signal on the INTR line persists. It also protects the ISP from being interrupted. When the 8086 is interrupted by a signal on the INTR line, the external hardware, which interrupted the 8086, has to supply the type number. On receiving a signal on the INTR line, the 8086 initiates two interrupt acknowledge machine cycles. During the first machine cycle, the 8086 sends an INTA signal to the external hardware (such as 8259A). This signal tells the external hardware to get ready to supply the type number. The 8086 then sends a second INTA signal to the external hardware, during the second machine cycle. On receiving the second INTA signal, the external hardware places the type number on the lower data bus AD0-AD7, and hand it over to the 8086. The 8086 then uses this type number to get the CS and IP values for the start of the ISP from the interrupt vector table. The ISP is then executed as a response to the interrupt.	7	7
OR			
4	Instructions to perform shift operations		

	• SHL/SAL – Used to shift bits of a byte/word towards left and put zero(S) in LSBs. • SHR – Used to shift bits of a byte/word towards the right and put zero(S) in MSBs. • SAR – Used to shift bits of a byte/word towards the right and copy the old MSB into the new MSB. Instructions to perform rotate operations • ROL – Used to rotate bits of byte/word towards the left, i.e. MSB to LSB and to Carry Flag [CF]. • ROR – Used to rotate bits of byte/word towards the right, i.e. LSB to MSB and to Carry Flag [CF]. • RCR – Used to rotate bits of byte/word towards the right, i.e. LSB to CF and CF to MSB. • RCL – Used to rotate bits of byte/word towards the left, i.e. MSB to CF and CF to LSB.	7	
5	<pre> DATA SEGMENT NUM_1 DW FA12H NUM_2 DB B5H RESULT DW 2DUP(0) DATA ENDS CODE SEGMENT ASSUME CS:CODE,DS:DATA MOV AX,DATA MOV DS,AX MOV AX,NUM_1 MOV BL,NUM_2 MOV BH,00H MUL BX MOV RESULT,AX MOV RESULT+2,DX CODE ENDS END </pre>	7	
OR			
6	‘N’ can be any number from 00H to FFH. Let it be ten (0AH). Take it as the count. <pre> DATA SEGMENT COUNT DB 0AH NUM DB 21H,67H,28H,12H,3AH,45H,ACH,2BH,57H,12H AVRG DB 1DUP(0) DATA ENDS CODE SEGMENT ASSUME CS:CODE,DS:DATA MOV AX,DATA MOV DS,AX MOV BX,OFFSET NUM MOV CL,COUNT MOV CH,00H MOV AX,0000H </pre>	7	7

	<pre> BACK:ADD AL,[BX] JNC NEXT INC AH NEXT:INC BX LOOP BACK DIV COUNT MOV AVR,AL CODE ENDS END </pre>		
7	Real Address Mode ▪ The 80386 processor will enter the real address mode when it is reset. ▪ This mode is similar to 8086 but allows access to the 32-bit registers. ▪ In real address mode the default operand size is 16-bits. ▪ In real address mode the processor can access only 1Mb of physical memory space like 8086 using 20-bit address. Therefore, the address lines A2-A19 alone are active. ▪ The 20-bit physical address is calculated by multiplying the segment register by 16_{10} and adding to an offset. ▪ In real address mode the size of the memory segments is 64kb. ▪ All the instructions of 80386 except a few can be executed in this mode. ▪ The primary purpose of real address mode is to set up the 80386 processor for protected virtual address mode of operation. Protected Virtual Address Mode (PVAM) ▪ The processor can switch from real address mode to PVAM by setting the PE bit (Protection Enable bit) in the control register CR0. ▪ In PVAM the processor can run the programs with sophisticated memory management and hardware assisted protection mechanism. ▪ In PVAM the processor has 4Gb of physical memory address space and 64Tb of virtual memory address space. ▪ The address lines A2-A31 along with four bank select signals BE0 to BE3 are used to address 4Gb of physical memory.	7	
OR			
8	• In this mode, the 80386 MMU operates similar to 80286. • Virtual addresses (48 bits long) are represented with a selector component and an offset component. • The selector component is used to index a descriptor in a descriptor table (LDT or GDT decided by the TI bit in the selector). • The descriptor contains the 32-bit physical base address for the segment. • The offset part of the virtual address is added to the base address to produce the actual physical address. • The offset part of a virtual address is 32 bits long so the segments can be as large as 4 gigabytes.		7

	Virtual Address physical memory Max size of a segment } 4 G bytes	Fig-4 Exp-3
9	- Complex instruction set with RISC features 32-bit, 2-issue, superscalar processor - Clock frequency – 60/66/90/100/133 MHz - Number of transistors – 3.1 million 32-bit address bus - Address range – $2^{32} = 4\text{Gb}$ 64-bit external data bus - On chip cache (L1) – 16Kb (8Kb data + 8Kb code) - External cache – 64 to 512Kb - Performance – 100 - 200 MIPS - Supports ‘Dynamic Branch Prediction Logic’ - On chip enhanced floating point unit. - Bus cycle pipelining - Address parity - Internal parity checking (Any seven)	7
10	OR The execution unit of Pentium processor has two parallel instruction pipelines, the "u" pipe and the "v" pipe with individual ALU for each pipe. The u-pipe can execute all integer and floating point instructions. The v-pipe can execute simple integer instructions and the FXCH floating-point instructions. While executing instructions, the processor checks next two	7

	instructions. If the execution of one instruction does not depend on the other then the first instruction is issued to the u-pipe and the second instruction is issued to the v-pipe, so that the two instructions can be executed simultaneously. If it is not possible to execute the instructions simultaneously, then the two instructions are issued to the u-pipe one by one and no instruction is issued to u-pipe.		
11	 The internal architecture of Intel Core 2 Duo is shown in figure. It is a dual-core processor with each core having the Core 2 microarchitecture. ➤ The cores are integrated on the same die. ➤ Each core has its own private 64-Kb level-1 cache (32-Kb instruction cache and 32-Kb data cache). ➤ It also has a shared 2 to 4-Mb level-2 cache. ➤ Each core has an advanced programmable interrupt controller that handles external interrupts as well as the inter-processor interrupts. ➤ The microarchitecture employs a 14-stage pipeline, 4 wide decode unit and 4 wide retirement unit. It also employs enhanced ALUs and deeper buffers. ➤ The decode unit has 32-byte pre-decode fetch buffer, 18 entry instruction queue and 4 decoders. One decoder is a complex one and the other three are simple. ➤ The three simple decoders deal with x86 instructions that map to a single uop, while the complex decoder handles instructions that produce 1-4 uops. ➤ The decoding take place under the control of a micro-code sequencer. ➤ The decoder has a special feature called macro-fusion. It allows two x86 instructions to be combined together and decoded to a single μop. Only one such fusion is permitted per cycle. This feature reduces μops by 10%.	Fig-3 Exp-4	

	➤ The execution unit has 6 dispatch ports and five ALUs; 3 integer ALUs and 2 floating point ALUs. ➤ It can consistently execute three instructions per cycle. ➤ Out of order execution, register renaming and deep branch prediction are used to increase the instruction throughput.		7
OR			
12	i) Interconnection: With multiple cores and shared resources such as memory, buses, and I/O, the interconnection between cores is a big issue. Bus contention and inter-processor communication are the main problems in any multiprocessing systems which is the same for multicore processors. ii) Cache Coherence: Cache coherence is an issue in multicore processors because of the distributed private caches and shared main memory. In most multicore processors, the L1-caches are private and the L2-caches are shared. Since each core has its own private cache, the copy of the data in that cache may not be the up-to-date version. For ensuring cache coherence, two protocols; snooping protocol and directory based protocols are used. iii) Programming aspects: Exploiting a multicore processor to its full potential is a big issue. Intelligent programming with multiple thread possibilities or parallelization is essential to ensure simultaneous execution in all the cores and thus fully exploiting the potentials of multicore processor. iv) Operating System Support: The operating system that used with a multicore processor must be able to handle the complexities of such a system to fully exploit the available resources.	7	

(6 × 7 = 42 Marks)

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